

Jiarui Hai

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Education

Johns Hopkins University

Ph.D Student in Electrical and Computer Engineering

Advisor: Prof. Mounya Elhilali

Baltimore, MD, USA

Aug 2022 - Present

Tsinghua University

M.E. in Civil Engineering

Member in Big Data Program

Beijing, China

Aug 2020 - Jun 2022

Tsinghua University

B.E. in Civil Engineering

B.S. in Business Analytics

Beijing, China

Aug 2016 - Jun 2020

Experience

Johns Hopkins University | LCAP

Research Assistant | Supervisor: Mounya Elhilali

Research Topic: Audio Generation and Understanding

Baltimore, MD, USA

Aug 2022 - Present

Apple

ML Research Intern

Research details confidential pending review

USA

Apr 2026 - Aug 2026

Adobe | CAVA

Research Scientist Intern | Supervisors: Ke Chen, Rithesh Kumar, Zeyu Jin

Research Topic: Speech Synthesis, Voice Design

San Francisco, CA, USA

May 2025 - Feb 2026

Tencent | AI Lab

Research Scientist Intern | Supervisors: Yong Xu, Hao Zhang, Dong Yu

Research Topic: Text-to-Audio Generation, Diffusion Models

Bellevue, WA, USA

May 2024 - Aug 2024

Kuaishou | AI Platform

Algorithm Engineer Intern | Supervisors: Rui Lu, Zhiyao Duan

Research Topic: Music Tagging, Singing Voice Synthesis

Beijing, China

Jun 2020 - Jan 2021

Selected Publications

* Equal Contribution † Research Mentorship

Full list available on Google Scholar

- [15] **Jiarui Hai**, Karan Thakkar, Ke Chen, Yunyun Wang, Jiaqi Su, Rithesh Kumar, Mounya Elhilali, Zeyu Jin. "VoiceDesigner: Text-to-Voice Generation and Editing via Unified Diffusion Modeling and Data Augmentation". Pre-print 2026.
- [14] Aaron Yee, Fengjie Lu, **Jiarui Hai**†, Chenang Jiang, Helin Wang, Siwei Tu, Weitao You, Lingyun Sun "VoiceTrace: A Benchmark and Retrieval Framework for Who-Said-What Speech Retrieval". Pre-print 2026.
- [13] Fengjie Lu, Chenang Jiang, **Jiarui Hai**†, Helin Wang, Aaron Yee "ALM2Vec: Learning Audio Embeddings for Universal Audio Retrieval with Large Audio-Language Models". Pre-print 2026.
- [12] Yongyi Zang, **Jiarui Hai**, Wanying Ge, Qiuqiang Kong, Zheqi Dai, Helin Wang, Yuki Mitsufuji, Mark D Plumbley. "Summary of The Inaugural Music Source Restoration Challenge". ICASSP 2026.
- [11] Helin Wang*, **Jiarui Hai***, Dading Chong, Karan Thakkar, Tiantian Feng, Dongchao Yang, Junhyeok Lee, Laureano Moro Velazquez, Jesus Villalba, Zengyi Qin, Shrikanth Narayanan, Mounya Elhilali, Najim Dehak. "CapSpeech: Enabling Downstream Applications in Style-Captioned Text-to-Speech". TASLP 2026.
- [10] Helin Wang, **Jiarui Hai**, Dongchao Yang, Chen Chen, Kai Li, Junyi Peng, Thomas Thebaud, Laureano Moro-Velazquez, Jesus Villalba, Najim Dehak. "SoloSpeech: Enhancing Intelligibility and Quality in Target Speech Extraction through a Cascaded Generative Pipeline". TASLP 2026.
- [9] **Jiarui Hai**, Helin Wang, Weizhe Guo, Mounya Elhilali. "FlexSED: Towards Open-Vocabulary Sound Event Detection". WASPAA 2025 (Oral).
- [8] **Jiarui Hai**, Mounya Elhilali. "SynSonic: Augmenting Sound Event Detection through Text-to-Audio Diffusion ControlNet and Effective Sample Filtering". WASPAA 2025.

- [7] **Jiarui Hai**, Yong Xu, Hao Zhang, Chenxing Li, Helin Wang, Mounya Elhilali, Dong Yu. "EzAudio: Enhancing Text-to-Audio Generation with Efficient Diffusion Transformer". Interspeech 2025 (**Oral**).
- [6] Helin Wang*, **Jiarui Hai***, Yen-Ju Lu, Karan Thakkar, Mounya Elhilali, Najim Dehak: "SoloAudio: Target Sound Extraction with Language-oriented Audio Diffusion Transformer". ICASSP 2025.
- [5] **Jiarui Hai***, Karan Thakkar*, Helin Wang, Zengyi Qin, Mounya Elhilali. "DreamVoice: Text-Guided Voice Conversion". Interspeech 2024.
- [4] **Jiarui Hai***, Helin Wang*, Dongchao Yang, Karan Thakkar, Najim Dehak, Mounya Elhilali. "DPM-TSE: A Diffusion Probabilistic Model for Target Sound Extraction". ICASSP 2024.
- [3] **Jiarui Hai***, Yu-jeh Liu*, Mounya Elhilali. "Boosting Modality Representation with Pre-trained Models and Multi-task Training for Multimodal Sentiment Analysis". ASRU 2023.
- [2] **Jiarui Hai**, Mounya Elhilali. "Diff-Pitcher: Diffusion-based Singing Voice Pitch Correction". WASPAA 2023 (**Oral**).
- [1] Rui Lu, Baigong Zheng, **Jiarui Hai**, Fei Tao, Zhiyao Duan, Ji Liu. "Progressive Teacher-Student Training Framework for Music Tagging". ICASSP 2022.

Research Activities

Datasets, Models, Benchmarks, and Challenges

- Music Source Restoration Challenge: A benchmark for music source restoration: <https://msrchallenge.com/>
- CapSpeech: A large-scale voice captioning dataset: <https://wanghelin1997.github.io/CapSpeech-demo/>
- DreamVoice: A voice timbre description dataset: https://haidog-yaqub.github.io/dreamvoice_demo/

Open-Source Repositories

- OpenSound: Built interactive demos for audio and speech models on Hugging Face: <https://huggingface.co/OpenSound>
- MeanFlow: PyTorch implementation of MeanFlow for diffusion accelerating : <https://github.com/haidog-yaqub/MeanFlow>
- EzAudio: Efficient diffusion transformer optimized for audio latents: <https://github.com/haidog-yaqub/EzAudio>

Peer Reviewer

- Conference: ICASSP, Interspeech, WASPAA, ASRU, SLT
- Journal: IJCV, TASP
- Workshop: DeLTa @ ICLR 2025, Audio Imagination @ NeurIPS 2024

Music Activities

Professional Recognition

- Featured Producer on a rap single showcased in HIPHOP BANK (TV show, China) | Aug 2021
- Invited Lecture on Music Production, Modern Sky (Chinas leading music company) | Jun 2021
- Top New Producer Award, BeatsHome Hip-hop Production Contest | Apr 2021

Creative & Media Activities

- Independent Artist & Beatmaker: produce singles and beat tapes
- Content Creator: share tutorials on music production and mixing

Leadership and Co-founder

- President, Rap & HipHop Association, Tsinghua University | 2020–2021
- Co-founder, InnerSup Records (Music Producer Team)
- Co-founder, Music Production Team, Rap & HipHop Association, Tsinghua University